

Exercises - III

- [1] Let $\mathbb{T}^n$ denote the n -dimensional torus which is an n -fold product of circles. Show that an even dimensional torus carries a symplectic form. Find two symplectic forms on $\mathbb{T}^2$ which are not isotopic but symplectomorphic.
- [2] Let (ω, J, g) be a compatible triple consisting of a symplectic form ω , a complex structure J and a Riemannian metric g on M . Let $f : M \rightarrow \mathbb{R}$ be a smooth function on M . Define the gradient the vector field ∇f of f by the relation $g(\nabla f, -) = df$. Find a relation between ∇f and X_f . Show that if f is a submersion then the flow lines of ∇f intersect the flow lines of X_f orthogonally.
- [3] Let C be a circle in the yz -plane with centre at $(a, 0)$ and radius $b < a$, and $\mathbb{T}^2$ be the surface of revolution in $\mathbb{R}^3$ obtained by revolving C about the z -axis. Consider the diffeomorphism $\varphi : \mathbb{T}^2 \rightarrow \mathbb{T}^2$ which is the restriction of a rotation about the z -axis. Note that φ has no fixed point. Does it violate Arnold's conjecture on number of fixed points?